

Topics for week 6

Chapter 4: System of Differential Equations

SDE

- 4.4 Classification of equilibrium points
- 4.5 Real distinct eigenvalues
- 4.6 Complex eigenvalues
- 4.7 Repeated eigenvalues

General Solution of $\mathbf{y}' = \mathbf{A}\mathbf{y}$

$\mathbf{A}$ (diagonalizable) $n \times n$ matrix

i. n linearly independent eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$

ii. n eigenvalues $r_1, r_2, \dots, r_n$ (may be repeated)

iii. Form n linearly independent solutions $\mathbf{v}_1 e^{r_1 t}, \mathbf{v}_2 e^{r_2 t}, \dots, \mathbf{v}_n e^{r_n t}$

vectors with functions as entries

iv. $\{\mathbf{v}_1 e^{r_1 t}, \mathbf{v}_2 e^{r_2 t}, \dots, \mathbf{v}_n e^{r_n t}\}$ is a fundamental set of solutions of the SDE

v. The general solution of the SDE is given by:

$$\mathbf{y} = c_1 \mathbf{v}_1 e^{r_1 t} + c_2 \mathbf{v}_2 e^{r_2 t} + \dots + c_n \mathbf{v}_n e^{r_n t}$$

$c_1, c_2, \dots, c_n$ can be any values (parameters)

$$\begin{pmatrix} 1 \\ 4 \\ -2 \\ 5 \end{pmatrix} e^{3t} = \begin{pmatrix} e^{3t} \\ 4e^{3t} \\ -2e^{3t} \\ 5e^{3t} \end{pmatrix}$$

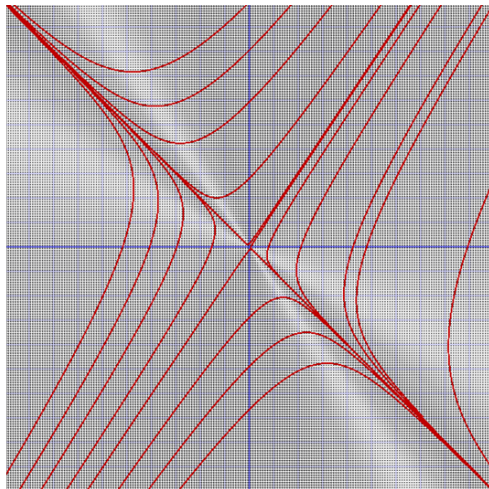
Trajectories and Phase Portraits

$$\begin{cases} y_1(t_0) = a \\ y_2(t_0) = b \end{cases}$$

$$\begin{cases} y_1' = y_1 + 2y_2 \\ y_2' = 3y_1 + 2y_2 \end{cases}$$

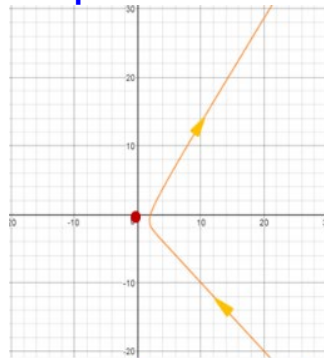
General Solution

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = c_1 \begin{pmatrix} 2 \\ 3 \end{pmatrix} e^{4t} + c_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{-t}$$

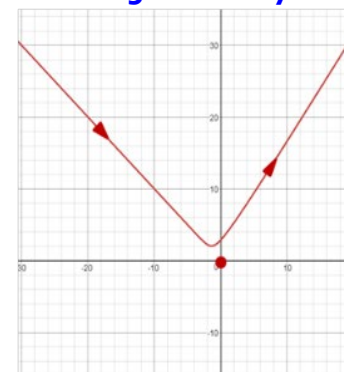


phase portrait of the system

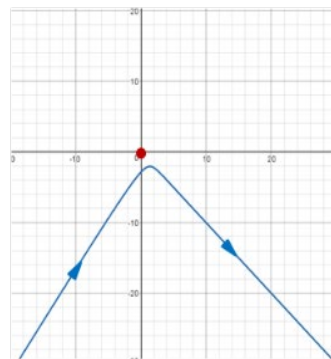
Each particular solution is represented by a trajectory



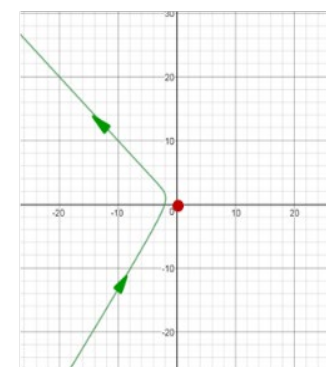
$c_1 = 1, c_2 = -1$



$c_1 = 1, c_2 = 1$

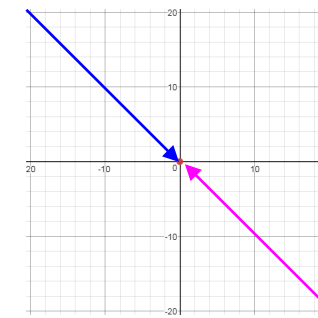


$c_1 = -1, c_2 = 1$

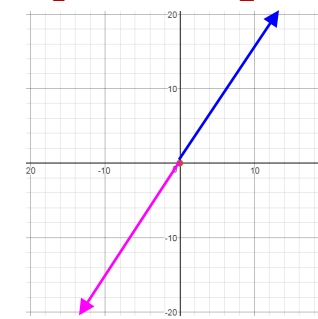


$c_1 = -1, c_2 = -1$

The initial condition is represented by a point on a particular trajectory



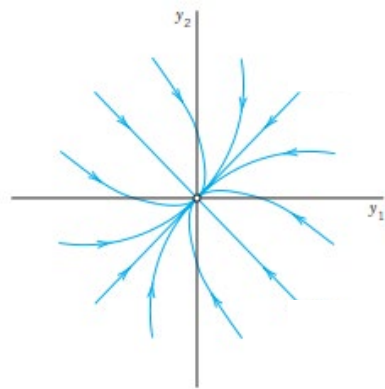
$c_1 = 0 \& c_2 \neq 0$



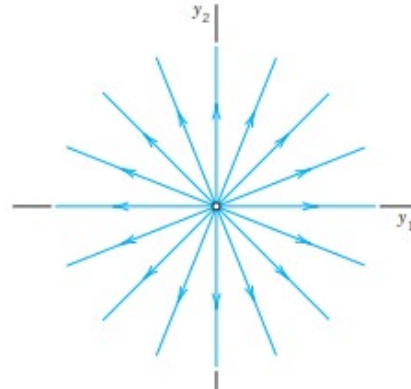
$c_1 \neq 0 \& c_2 = 0$

$c_1 = 0 \& c_2 = 0$
No trajectory
equilibrium
point

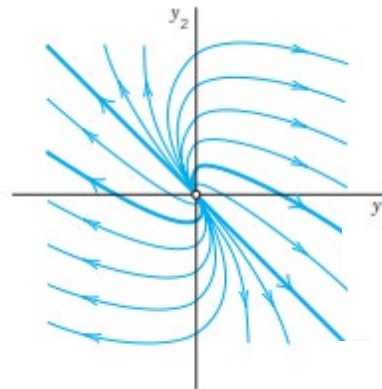
Types of Equilibrium Points



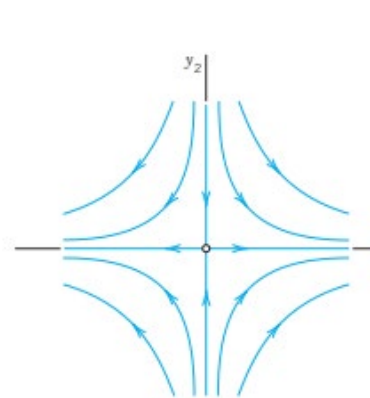
Node



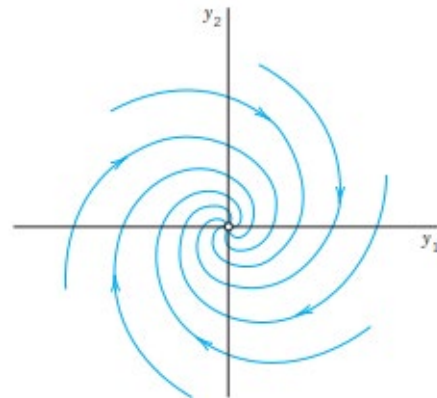
Star Node



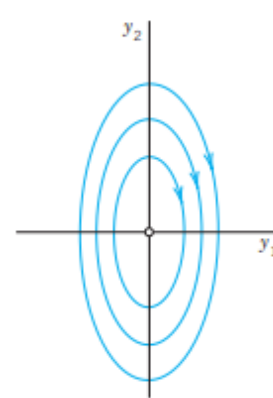
Degenerate Node



Saddle Point



Spiral Point



Center

Stability of e.p.

- arrow pointing away from origin
 - Unstable
- arrow pointing towards origin
 - Asymptotically stable
- trajectories circling around origin
 - Neutrally stable

Eigenvalue Criteria

2 x 2 matrix **A**

Eigenvalues r_1 and r_2	Equilibrium Point	Stablility
Real & distinct (both positive)	Node	Unstable
Real & distinct (both negative)	Node	Asymptotically stable
Real & repeated (positive) (2 indep. eigenvectors)	Star node	Unstable
Real & repeated (negative) (2 indep. eigenvectors)	Star node	Asymptotically stable
Real & repeated (positive) (1 indep. eigenvector)	Degenerate node	Unstable
Real & repeated (negative) (1 indep. eigenvector)	Degenerate node	Asymptotically stable
Real (opposite sign)	Saddle point	Unstable
Complex (positive real parts)	Spiral point	Unstable
Complex (negative real parts)	Spiral point	Asymptotically stable
Pure imaginary	Center	Neutrally stable

$$\begin{cases} y_1' &= 5y_1 - y_2 \\ y_2' &= -4y_1 + 2y_2 \end{cases} \quad \mathbf{A} = \begin{pmatrix} 5 & -1 \\ -4 & 2 \end{pmatrix}$$

Example (node)

- Find eigenvalues and eigenvectors
- Find fundamental set of solutions
- Write down general solution
- IVP: Find particular solution
- Equilibrium point (type and stability)
- Sketch trajectories

Eigenvalues $r_1 = 1$ and $r_2 = 6$

Eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$ for r_1 , and $\mathbf{v}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ for r_2

Fundamental set of solutions

$$\mathbf{x}_1 = \begin{pmatrix} 1 \\ 4 \end{pmatrix} e^t \quad \mathbf{x}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{6t}$$

General Solution

$$\mathbf{y} = c_1 \begin{pmatrix} e^t \\ 4e^t \end{pmatrix} + c_2 \begin{pmatrix} -e^{6t} \\ e^{6t} \end{pmatrix}$$

$$\begin{cases} y_1' &= 5y_1 - y_2 \\ y_2' &= -4y_1 + 2y_2 \end{cases} \quad \mathbf{A} = \begin{pmatrix} 5 & -1 \\ -4 & 2 \end{pmatrix}$$

Example (node)

- Find eigenvalues and eigenvectors
- Find fundamental set of solutions
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- IVP: Find particular solution
- Equilibrium point (type and stability)
- Sketch trajectories

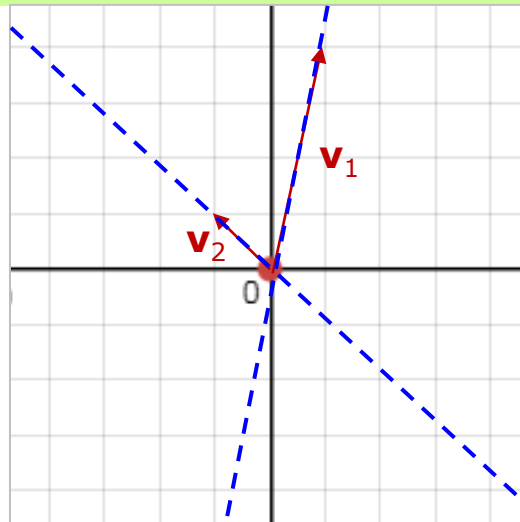
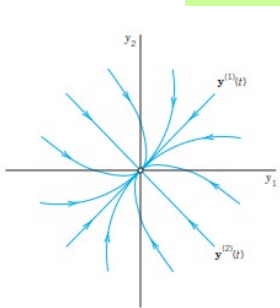
Eigenvalues $r_1 = 1$ and $r_2 = 6$

$$r_1 < r_2$$

Eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$ for r_1 , and $\mathbf{v}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ for r_2

real and distinct, both positive

→ equilibrium point: Node, unstable



start near origin and parallel to $\mathbf{v}_1$ (smaller eigenvalues r_1)

move away from origin and parallel to $\mathbf{v}_2$ (larger eigenvalues r_2)

Directions of trajectories

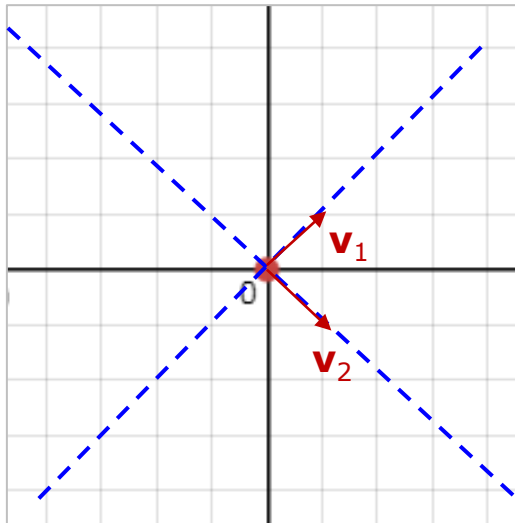
$$\mathbf{y}' = \mathbf{A}\mathbf{y}$$

Eigenvalues $r_1 = -3$ and $r_2 = -2$

Eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ for r_1 , and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ for r_2

Eigenvalues: real and distinct $\rightarrow$ **Node**

Eigenvalues: both negative $\rightarrow$ **Asymptotically stable**



start (far) away from origin and parallel to $\mathbf{v}_1$ (smaller eigenvalues r_1)

move towards origin and parallel to $\mathbf{v}_2$ (larger eigenvalues r_2)

$$\begin{cases} y_1' &= -y_1 - 2y_2 \\ y_2' &= -3y_1 - 2y_2 \end{cases} \quad \mathbf{A} = \begin{pmatrix} -1 & -2 \\ -3 & -2 \end{pmatrix}$$

Example (saddle point)

- Find eigenvalues and eigenvectors
- Find fundamental set of solutions
- Write down general solution
- IVP: Find particular solution
- Equilibrium point (type and stability)
- Sketch trajectories

Eigenvalues $r_1 = -4$ and $r_2 = 1$

Eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ for r_1 , and $\mathbf{v}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ for r_2

Fundamental set of solutions

$$\mathbf{x}_1 = \begin{pmatrix} 2 \\ 3 \end{pmatrix} e^{-4t} \quad \mathbf{x}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^t$$

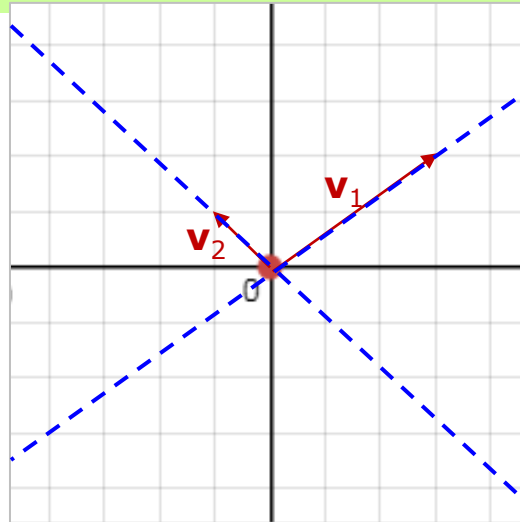
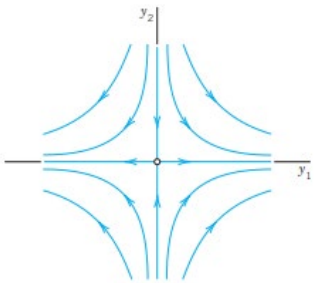
General Solution

$$\mathbf{y} = c_1 \begin{pmatrix} 2e^{-4t} \\ 3e^{-4t} \end{pmatrix} + c_2 \begin{pmatrix} -e^t \\ e^t \end{pmatrix}$$

$$\begin{cases} y_1' &= -y_1 - 2y_2 \\ y_2' &= -3y_1 - 2y_2 \end{cases} \quad \mathbf{A} = \begin{pmatrix} -1 & -2 \\ -3 & -2 \end{pmatrix}$$

Example (saddle point)

- Find eigenvalues and eigenvectors
- Find fundamental set of solutions
- Write down general solution
- IVP: Find particular solution
- Equilibrium point (type and stability)
- Sketch trajectories



Eigenvalues $r_1 = -4$ and $r_2 = 1$ $r_1 < r_2$

Eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ for r_1 , and $\mathbf{v}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ for r_2

real and distinct, opposite sign

→ equilibrium point: saddle point, unstable

Start parallel to $\mathbf{v}_1$ and moving towards origin

End parallel to $\mathbf{v}_2$ and moving away from origin

Complex Eigenvalues

Eigenvalues $r_1 = a + b i$ and $r_2 = a - b i$ (complex conjugates)

Eigenvectors $\mathbf{v}_1 = \begin{pmatrix} c + d i \\ e + f i \end{pmatrix} = \mathbf{p} + \mathbf{q} i$ and $\mathbf{v}_2 = \begin{pmatrix} c - d i \\ e - f i \end{pmatrix} = \mathbf{p} - \mathbf{q} i$

Complex solutions

$$\mathbf{x}_1(t) = \mathbf{v}_1 e^{r_1 t}$$

$$\mathbf{x}_2(t) = \mathbf{v}_2 e^{r_2 t}$$

Real solutions

$$\mathbf{x}_{\text{Re}}(t) = e^{at}(\mathbf{p} \cos bt - \mathbf{q} \sin bt)$$

$$\mathbf{x}_{\text{Im}}(t) = e^{at}(\mathbf{p} \sin bt + \mathbf{q} \cos bt)$$

General solution

$$\mathbf{y} = c_1 \mathbf{x}_{\text{Re}}(t) + c_2 \mathbf{x}_{\text{Im}}(t)$$

$$\begin{cases} y_1' &= 2y_1 - 4y_2 \\ y_2' &= 4y_1 + 2y_2 \end{cases} \quad \mathbf{A} = \begin{pmatrix} 2 & -4 \\ 4 & 2 \end{pmatrix}$$

Example (spiral point)

- Find eigenvalues and eigenvectors
- Find fundamental set of **real** solutions
- Write down general solution
- IVP: Find particular solution
- Equilibrium point (type and stability)
- Sketch trajectories

Using MATLAB

```
>> [P D]=eig(sym(A))
P =
[-1i, 1i]
[ 1, 1]
D =
[2 - 4i, 0]
[ 0, 2 + 4i]
```

Eigenvalues $r_1 = 2 + 4i$ $r_2 = 2 - 4i$

Eigenvectors $\mathbf{v}_1 = \begin{pmatrix} i \\ 1 \end{pmatrix}$ $\mathbf{v}_2 = \begin{pmatrix} -i \\ 1 \end{pmatrix}$

$$\begin{cases} y_1' &= 2y_1 - 4y_2 \\ y_2' &= 4y_1 + 2y_2 \end{cases} \quad \mathbf{A} = \begin{pmatrix} 2 & -4 \\ 4 & 2 \end{pmatrix}$$

Example (spiral point)

- Find eigenvalues and eigenvectors
- Find fundamental set of **real** solutions
- Write down general solution
- IVP: Find particular solution
- Equilibrium point (type and stability)
- Sketch trajectories

Eigenvalues $r_1 = 2 + 4i$, $r_2 = 2 - 4i$

$$a = 2, b = 4$$

Eigenvector $\mathbf{v}_1 = \begin{pmatrix} i \\ 1 \end{pmatrix}$ $\mathbf{v}_2 = \begin{pmatrix} -i \\ 1 \end{pmatrix}$

$$\mathbf{p} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \mathbf{q} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Fundamental set of (real) solutions

$$\mathbf{x}_{\text{Re}}(t) = e^{at}(\mathbf{p} \cos bt - \mathbf{q} \sin bt) = e^{2t} \left(\begin{pmatrix} 0 \\ 1 \end{pmatrix} \cos 4t - \begin{pmatrix} 1 \\ 0 \end{pmatrix} \sin 4t \right)$$

$$\mathbf{x}_{\text{Im}}(t) = e^{at}(\mathbf{p} \sin bt + \mathbf{q} \cos bt) = e^{2t} \left(\begin{pmatrix} 0 \\ 1 \end{pmatrix} \sin 4t + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cos 4t \right)$$

General solution

$$\mathbf{y} = c_1 \mathbf{x}_{\text{Re}}(t) + c_2 \mathbf{x}_{\text{Im}}(t) = e^{2t} \left[c_1 \begin{pmatrix} -\sin(4t) \\ \cos(4t) \end{pmatrix} + c_2 \begin{pmatrix} \cos(4t) \\ \sin(4t) \end{pmatrix} \right]$$

$$\begin{cases} y_1' &= 2y_1 - 4y_2 \\ y_2' &= 4y_1 + 2y_2 \end{cases} \quad \mathbf{A} = \begin{pmatrix} 2 & -4 \\ 4 & 2 \end{pmatrix}$$

Example (spiral point)

- Find eigenvalues and eigenvectors
- Find fundamental set of **real** solutions
- Write down general solution
- IVP: Find particular solution
- Equilibrium point (type and stability)
- Sketch trajectories

Eigenvalues $r_1 = 2 + 4i$, $r_2 = 2 - 4i$

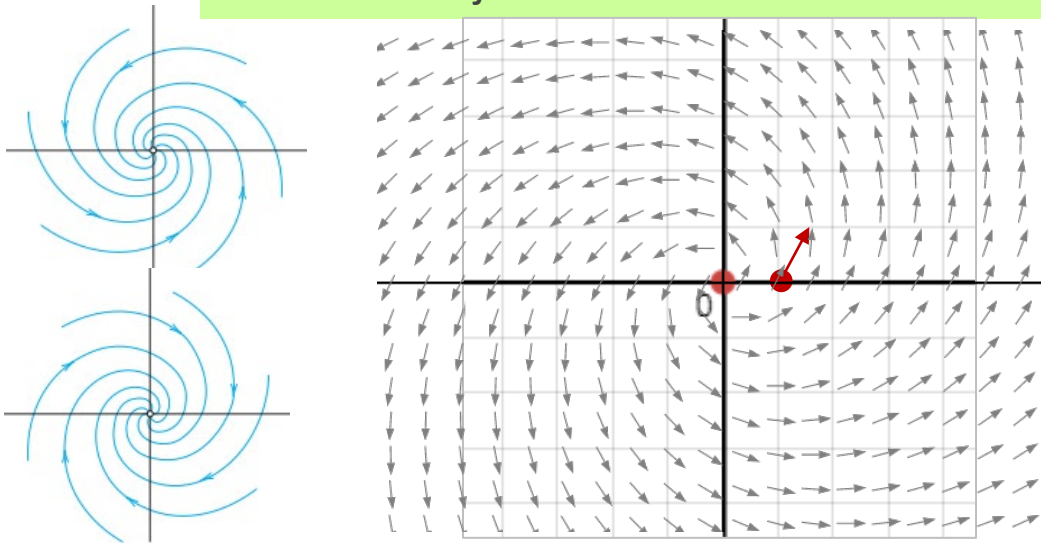
Eigenvector $\mathbf{v}_1 = \begin{pmatrix} i \\ 1 \end{pmatrix}$ $\mathbf{v}_2 = \begin{pmatrix} -i \\ 1 \end{pmatrix}$

Complex, real part > 0

→ **equilibrium point: spiral point, unstable**

Orientation of trajectories

- Test a point, say $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$
- Multiply by $\mathbf{A}$ to get $\begin{pmatrix} 2 & -4 \\ 4 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$
- Tangent vector $\mathbf{y}'$ at $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is $\begin{pmatrix} 2 \\ 4 \end{pmatrix}$, **pointing upward**



$$\begin{cases} y_1' &= 2y_1 + 4y_2 \\ y_2' &= -2y_1 - 2y_2 \end{cases} \quad \mathbf{A} = \begin{pmatrix} 2 & 4 \\ -2 & -2 \end{pmatrix}$$

Example (center)

- Find eigenvalues and eigenvectors
- Find fundamental set of **real** solutions
- Write down general solution
- IVP: Find particular solution
- Equilibrium point (type and stability)
- Sketch trajectories

Using MATLAB

```
>> [P D]=eig(sym(A))
P =
[- 1 + 1i, - 1 - 1i]
[      1,      1]
D =
[-2i, 0]
[ 0, 2i]
```

Eigenvalues $r_1 = 2i$ $r_2 = -2i$

Eigenvectors $\mathbf{v}_1 = \begin{pmatrix} -1 - i \\ 1 \end{pmatrix}$ $\mathbf{v}_2 = \begin{pmatrix} -1 + i \\ 1 \end{pmatrix}$

$$\begin{cases} y_1' &= 2y_1 + 4y_2 \\ y_2' &= -2y_1 - 2y_2 \end{cases} \quad \mathbf{A} = \begin{pmatrix} 2 & 4 \\ -2 & -2 \end{pmatrix}$$

Example (center)

- Find eigenvalues and eigenvectors
- Find fundamental set of **real** solutions
- Write down general solution
- IVP: Find particular solution
- Equilibrium point (type and stability)
- Sketch trajectories

Eigenvalues $r_1 = 2i, r_2 = -2i$

$$a = 0, b = 2$$

Eigenvector $\mathbf{v}_1 = \begin{pmatrix} -1 - i \\ 1 \end{pmatrix} \quad \mathbf{v}_2 = \begin{pmatrix} -1 + i \\ 1 \end{pmatrix}$

$$\mathbf{p} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \mathbf{q} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

purely imaginary (real part = 0)

→ **equilibrium point: center, neutrally stable**

Fundamental set of (real) solutions

$$\mathbf{x}_{\text{Re}}(t) = e^{at}(\mathbf{p} \cos bt - \mathbf{q} \sin bt) = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \cos 2t - \begin{pmatrix} -1 \\ 0 \end{pmatrix} \sin 2t$$

$$\mathbf{x}_{\text{Im}}(t) = e^{at}(\mathbf{p} \sin bt + \mathbf{q} \cos bt) = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \sin 2t + \begin{pmatrix} -1 \\ 0 \end{pmatrix} \cos 2t$$

What is the orientation of the trajectories?

General solution

$$\mathbf{y} = c_1 \mathbf{x}_{\text{Re}}(t) + c_2 \mathbf{x}_{\text{Im}}(t) = \left[c_1 \begin{pmatrix} -\cos(2t) + \sin(2t) \\ \cos(2t) \end{pmatrix} + c_2 \begin{pmatrix} -\sin(2t) - \cos(2t) \\ \sin(2t) \end{pmatrix} \right]$$

$$\begin{cases} y_1' &= 2y_1 + 4y_2 \\ y_2' &= -2y_1 - 2y_2 \end{cases} \quad \mathbf{A} = \begin{pmatrix} 2 & 4 \\ -2 & -2 \end{pmatrix}$$

Example (center)

- Find eigenvalues and eigenvectors
- Find fundamental set of **real** solutions
- Write down general solution
- IVP: Find particular solution
- Equilibrium point (type and stability)
- Sketch trajectories

Eigenvalues $r_1 = 2i, r_2 = -2i$

$$a = 0, b = 2$$

Eigenvector $\mathbf{v}_1 = \begin{pmatrix} -i & 1 \\ 1 & 1 \end{pmatrix} \quad \mathbf{v}_2 = \begin{pmatrix} i & 1 \\ 1 & 1 \end{pmatrix}$

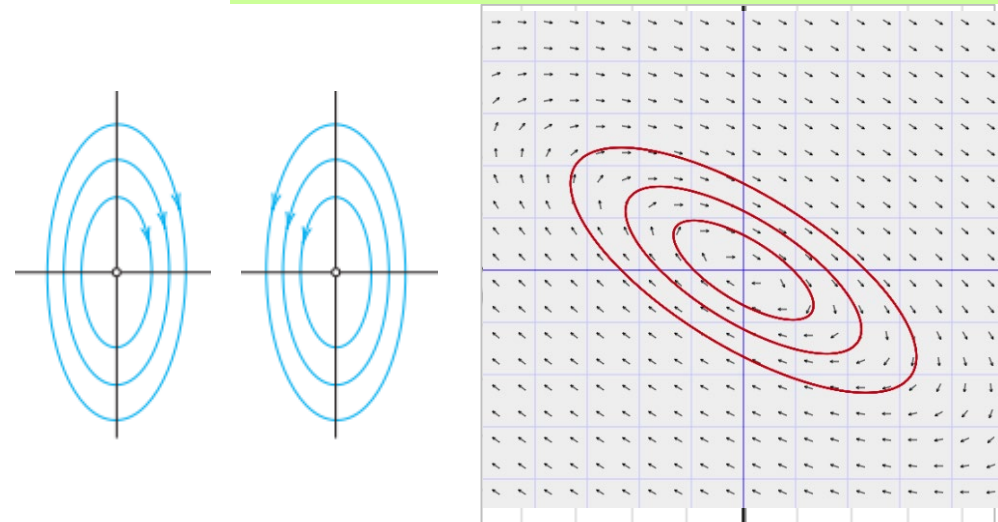
$$\mathbf{p} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \mathbf{q} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

purely imaginary (real part = 0)

→ **equilibrium point: center, neutrally stable**

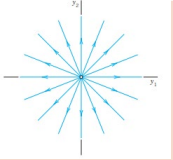
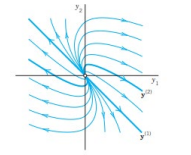
Orientation of trajectories

- Test a point, say $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$
- Multiply by $\mathbf{A}$ to get $\begin{pmatrix} 2 & 4 \\ -2 & -2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \end{pmatrix}$
- Tangent vector $\mathbf{y}'$ at $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is $\begin{pmatrix} 2 \\ -2 \end{pmatrix}$, **pointing downward**



$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Repeated Eigenvalue

A (2x2) eigenvalue r	Diagonalizable (scalar matrix)	Not diagonalizable
eigenvectors	2 linearly independent $\mathbf{v}_1, \mathbf{v}_2$ (every non-zero vector is eigenvector)	1 linearly independent $\mathbf{v}$
Equilibrium point	Star node 	Degenerate node 
Stability	Unstable ($r > 0$) Asymptotically stable ($r < 0$)	Unstable ($r > 0$) Asymptotically stable ($r < 0$)
General solution	$\mathbf{y} = c_1 \begin{pmatrix} e^{rt} \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ e^{rt} \end{pmatrix}$	$\mathbf{y} = c_1 e^{rt} \mathbf{v} + c_2 e^{rt} (t\mathbf{v} + \mathbf{u})$ $\mathbf{u}$: generalized eigenvector

Solve: $(\mathbf{A} - r\mathbf{I})\mathbf{u} = \mathbf{v}$

$$\begin{cases} y_1' &= 2y_1 \\ y_2' &= y_1 + 2y_2 \end{cases} \quad \mathbf{A} = \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}$$

Example (Degenerate node)

- Find eigenvalues and eigenvectors
- Find a **generalised** eigenvector
- Find fundamental set of solutions
- Write down general solution
- IVP: Find particular solution
- Equilibrium point (type and stability)
- Sketch trajectories

Fundamental set of solutions

$$\mathbf{x}_1(t) = e^{2t} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\mathbf{x}_2(t) = te^{2t} \mathbf{v} + e^{2t} \mathbf{u} = te^{2t} \begin{pmatrix} 0 \\ 1 \end{pmatrix} + e^{2t} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = e^{2t} \begin{pmatrix} 1 \\ t \end{pmatrix}$$

General solution

$$\mathbf{y} = c_1 e^{2t} \begin{pmatrix} 0 \\ 1 \end{pmatrix} + c_2 e^{2t} \begin{pmatrix} 1 \\ t \end{pmatrix}$$

Eigenvalues $r = 2$ (repeated)

Eigenvector $\mathbf{v} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ for r

Generalised eigenvector $\mathbf{u}$

Solve: $(\mathbf{A} - r\mathbf{I})\mathbf{u} = \mathbf{v}$

$$\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Get solution $\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 1 \\ s \end{pmatrix}$ s is a parameter

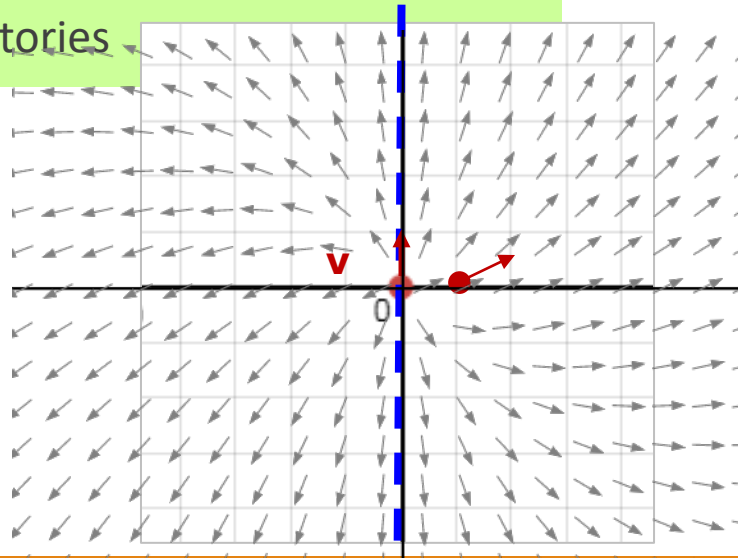
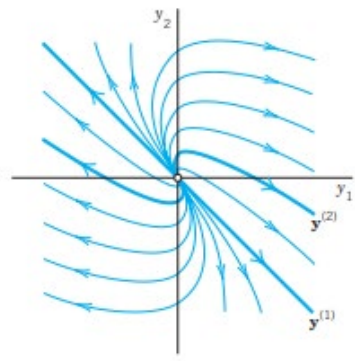
Choose $s = 0$ for a specific solution $\mathbf{u} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

You can choose other values of s to get other generalized eigenvectors $\mathbf{u}$

$$\begin{cases} y_1' &= 2y_1 \\ y_2' &= y_1 + 2y_2 \end{cases} \quad \mathbf{A} = \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}$$

Example (Degenerate node)

- Find eigenvalues and eigenvectors
- Find a **generalised** eigenvector
- Find fundamental set of solutions
- Write down general solution
- IVP: Find particular solution
- Equilibrium point (type and stability)
- Sketch trajectories



Eigenvalues $r = 2$ (repeated)

Eigenvector $\mathbf{v} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ for r

1 positive eigenvalue, only 1 indep. eigenvector
 $\rightarrow$ **equilibrium point: degenerate node, unstable**

Trajectories on opposite sides of the eigenvector move in opposite directions

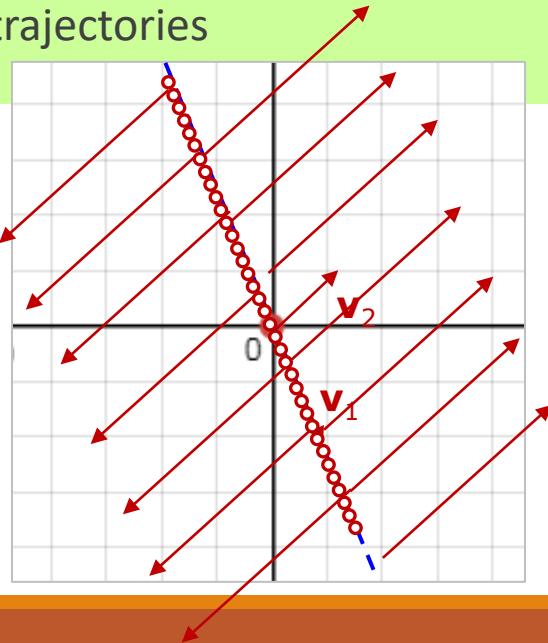
Orientation of trajectories

- Test a point, say $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$
- Multiply by $\mathbf{A}$ to get $\begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$
- Tangent vector $\mathbf{y}'$ at $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$, **pointing upward**

$$\begin{cases} y_1' &= 2y_1 + y_2 \\ y_2' &= 2y_1 + y_2 \end{cases} \quad \mathbf{A} = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix}$$

Example (0 eigenvalue)

- Find eigenvalues and eigenvectors
- Find fundamental set of solutions
- Write down general solution
- IVP: Find particular solution
- Equilibrium points (type and stability)
- Sketch trajectories



Eigenvalues $r_1 = 0$ and $r_2 = 3$

Eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$ for r_1 , and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ for r_2

Fundamental set of solutions

$$\mathbf{x}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix} \quad \mathbf{x}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{3t}$$

General Solution

$$\mathbf{y} = c_1 \begin{pmatrix} 1 \\ -2 \end{pmatrix} + c_2 \begin{pmatrix} e^{3t} \\ e^{3t} \end{pmatrix}$$